GENOMICS LABORATORY SEQUENCING WORKFLOW MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the Degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



EASWARI ENGINEERING COLLEGE
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OCTOBER 2026

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ABSTRACT

Next-Generation Sequencing (NGS) has emerged as a cornerstone of modern precision oncology and personalized medicine, enabling high-resolution molecular tumor profiling and targeted therapeutic intervention. However, molecular diagnostic laboratories encounter profound operational hurdles across high-volume sequencing operations, including fragmented manual accessioning, error-prone specimen tracking across physical freezer matrices, disparate quality control instrument outputs, and non-deterministic laboratory status handoffs. This project presents the Genomics Laboratory Sequencing Workflow Management System (Genomics LIMS), an enterprise-grade, full-stack software platform engineered to deliver comprehensive end-to-end traceability, automated instrument telemetry parsing, intelligent flow cell batching, and regulatory compliance under ISO 15189:2022 and 21 CFR Part 11 standards.

The platform features an automated accessioning engine generating standardized 2D DataMatrix barcoded identifiers, enforcing stringent pre-analytical metadata capture and physical rack coordinate mapping. To mitigate manual transcription errors, an automated Quality Control (QC) Telemetry Ingestion engine was developed to parse tabular and XML metrics exported directly from Agilent 4200 TapeStation and Qubit 4 Fluorometer instruments, extracting DNA Integrity Numbers (DIN), RNA Integrity Numbers (RIN), and fluorometric concentrations with automated gating thresholds. An intelligent sequencing batch optimizer multiplexes dual-indexed patient libraries onto Illumina NovaSeq and NextSeq flow cells (SP, S1, S2 configurations), verifying index compatibility and preventing index collision before sequencing execution.

The software integrates a 6-stage deterministic state transition pipeline spanning Accessioning, Library Preparation, Sequencing Execution, Secondary Bioinformatic Analysis, Somatic Variant Interpretation (AMP/ASCO/CAP tiers I-IV), and Sign-out. Furthermore, the system incorporates an HL7 FHIR Genomics JSON serialization engine that packages patient demographics, molecular findings, and pathogenic variant classifications into standardized FHIR bundles for seamless interoperability with hospital electronic health records (EHR). Complete cryptographic SHA-256 audit trails ensure regulatory accountability. Performance benchmarking demonstrates sub-200ms API response latency and an elimination of specimen mix-ups across extensive simulated clinical runs.

Keywords: *Genomics LIMS, Next-Generation Sequencing, Quality Control Telemetry, Automated Batching, Somatic Variant Classification, HL7 FHIR Interoperability, ISO 15189 Compliance, MERN Stack.*

ACKNOWLEDGEMENT

We express our sincere gratitude to the Principal Dr. P. Deiva Sundari and the Management of our college for providing us with the infrastructural facilities and computational resources to carry out this project work.

We convey our heartfelt thanks to the Head of the Department Dr. T. KalaiSelvi, Department of Artificial Intelligence and Data Science, for the continuous encouragement and support throughout this endeavor.

We are extremely thankful to our Project Guide Ms. D. S. K. Harshitha, Department of Artificial Intelligence and Data Science, for valuable guidance, insightful suggestions, and constant motivation throughout the completion of this project.

We would also like to extend our gratitude to all the faculty members of the Department of Artificial Intelligence and Data Science for their cooperation and constructive feedback.

Finally, we thank our parents and friends for their constant encouragement and help during the course of this project.

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CHAPTER 1
INTRODUCTION

This chapter deals with the overview of the Genomics Laboratory Sequencing Workflow Management System, its problem definition, objectives, scope, and the motivation behind developing this platform.

1.1 OVERVIEW

The clinical genomics and molecular diagnostics sector has witnessed exponential growth over the past decade, driven by the rise of precision medicine, Next-Generation Sequencing (NGS), and the increasing demand for high-throughput somatic mutation profiling in oncology. Commercial sequencers such as Illumina NovaSeq and NextSeq process hundreds of patient libraries simultaneously. However, diagnostic laboratories suffer from significant operational inefficiencies in pre-analytical accessioning, disparate quality control (QC) instrument outputs, and non-deterministic workflow handoffs, leading to suboptimal turnaround times for patient therapy selection.

The Genomics Laboratory Sequencing Workflow Management System addresses these limitations by introducing an intelligent, centralized platform that automates specimen tracking, microfluidic telemetry parsing, and dual-index multiplex batching. The system integrates real-time state broadcasting powered by WebSockets, structured milestone tracking across 6 clinical stages, and standardized HL7 FHIR Genomics JSON serialization for seamless electronic health record (EHR) interoperability. The platform is built using a modern, scalable technology stack comprising React with Vite for the frontend, Node.js with Express for the backend, MongoDB for document storage, Socket.IO for real-time telemetry events, and cryptographic audit trail logging.

By combining automated QC gating with immutable audit trails, the platform aims to reduce diagnostic turnaround time by 48 percent while eliminating specimen mix-ups and transcription errors. The architecture is designed to handle concurrent clinical users and real-time instrument telemetry feeds, making it suitable for production hospital deployments. JWT-based middleware manages authentication and role-based access control, ensuring a secure and performant experience for all laboratory personnel.

The platform serves molecular biologists, bioinformaticians, pathologists, and laboratory directors seeking zero-defect chain of custody. Smart quality gating is executed using automated TapeStation DIN and Qubit fluorometric telemetry ingestion, replacing manual spreadsheet tracking.

automated telemetry parsing and multi-factor validation techniques that analyze nucleic acid integrity, sample purity, and historical run performance data. This eliminates the manual, error-prone spreadsheet review that plagues existing laboratory platforms.

The platform also introduces milestone-based sample tracking through an immutable audit trail, where specimens advance only upon verified milestone completion. This mechanism significantly reduces sample rejection rates and builds trust between clinicians and laboratory staff. The combination of automated QC parsing, real-time collaboration, and regulatory compliance creates a comprehensive ecosystem for successful genomic workflows.

1.2 PROBLEM DEFINITION

Clinical personnel on existing laboratory platforms are overwhelmed by hundreds of manual data entry steps for every sequencing run. The lack of automated telemetry ingestion and intelligent sample tracking forces technicians to manually review and transcribe QC results, consuming significant time and effort. This inefficiency is compounded by the absence of standardized quality indicators, making it difficult to distinguish between genuinely high-integrity libraries and degraded specimens.

Simultaneously, urgent oncology samples struggle to maintain rapid turnaround times on crowded sequencing queues where static spreadsheet tracking dominates. Critical diagnostic specimens are often delayed behind routine research pools, leading to prolonged turnaround times that delay targeted cancer therapy initiation. This creates a frustrating experience where patient outcomes are compromised by laboratory bottlenecks.

The absence of integrated laboratory tracking tools forces clinical teams to rely on external spreadsheets, email threads, and paper logs for communication, file sharing, and progress monitoring. This fragmentation leads to miscommunication, sample mix-ups, and run failures. Milestone definitions are often ambiguous, and there is no automated system to verify sequencing readiness against agreed-upon criteria.

Compliance tracking represents another critical challenge in molecular diagnostics. Without tamper-evident audit trails compliant with ISO 15189:2022 and FDA 21 CFR Part 11, laboratories risk regulatory citations and compromised data integrity. The lack of a structured, automated audit mechanism further erodes trust and discourages seamless accreditation.

Therefore, there is a pressing need for a software-based Genomics LIMS that provides automated instrument telemetry parsing, integrated real-time collaboration tools, structured milestone tracking, and standardized HL7 FHIR report generation. Such a platform would significantly reduce turnaround times and ensure successful diagnostic outcomes for all patients.

1.3 OBJECTIVE

The primary objective of this project is to design and develop an automated quality control parsing engine that intelligently evaluates nucleic acid integrity based on DIN, RIN, concentration levels, and historical performance metrics. The parsing algorithm extracts tabular telemetry from Agilent TapeStation and Qubit fluorometer files, evaluating metric thresholds to ensure only qualified libraries advance to sequencing execution.

The second objective is to develop an integrated real-time notification and tracking system using WebSocket technology. This system enables instant communication between laboratory staff, alerts technicians of QC failures, and maintains a complete event history linked to specific specimen milestones. The goal is to eliminate the need for external communication tools and centralize all workflow interactions within the platform.

The third objective focuses on implementing a comprehensive milestone-based specimen tracking system. This module allows technicians and bioinformaticians to monitor 6 deterministic stages: Accessioned, Library Prep, Sequencing Execution, Bioinformatic Analysis, Clinical Review, and Signed Out. Automated progress monitoring tracks milestone completion and triggers notifications, ensuring transparency and accountability.

The fourth objective involves creating an intelligent flow cell batching optimizer that multiplexes dual-indexed patient libraries onto Illumina NovaSeq (SP, S1, S2) flow cells while performing automated collision checks on i5/i7 adapter sequences. This mechanism eliminates index transposition errors and maximizes flow cell throughput efficiency.

The fifth objective is to design an HL7 FHIR Genomics serialization engine that exports clinical diagnostic reports as standard JSON bundles. Reports encapsulate somatic variant classifications (AMP/ASCO/CAP Tiers I-IV) and pathogenic HGVS nomenclature, enabling direct electronic health record ingestion.

along with cryptographic SHA-256 audit trails and role-based permissions to guarantee regulatory compliance under ISO 15189 and 21 CFR Part 11.

1.4 SCOPE OF THE PROJECT

The platform targets molecular diagnostic laboratories, hospital pathology core facilities, and contract clinical research organizations operating in oncology, rare disease genetics, and precision therapeutics. It supports workflow-based operations ranging from individual STAT biopsy specimens to high-throughput multi-flow cell production runs, accommodating diverse sequencing panels including targeted hotspot panels, whole-exome sequencing, and comprehensive genomic profiling.

The core functionality encompasses digitized sample accessioning with 2D DataMatrix barcodes, automated Agilent TapeStation CSV parsing, real-time milestone tracking, flow cell batching with dual-index verification, somatic variant classification under AMP/ASCO/CAP guidelines, and HL7 FHIR report generation. The platform provides dedicated dashboards for accessioning technicians, molecular biologists, bioinformaticians, pathologists, and laboratory directors.

From a technical perspective, the scope includes the development of RESTful APIs using Express.js, real-time event processing with Socket.IO for instant telemetry alerts, CSV parsing microservices for instrument log ingestion, and JWT middleware for authentication and role-based access.

The application follows a monorepo structure with separate client and server directories, using Vite for frontend bundling and Nodemon for backend hot-reloading during development. The system is designed for straightforward deployment on cloud platforms or local intranet servers, supporting a growing laboratory workload with minimal configuration overhead.

Certain aspects are explicitly excluded from the initial project scope. These include direct physical robotic liquid handler firmware control (such as Hamilton Microlab STAR APIs), physical sequencer laser calibration, and live hospital EHR integration over dedicated MLLP VPN tunnels requiring institutional IRB ethics review. These features are identified as potential future enhancements based on laboratory adoption.

1.5 MOTIVATION

The motivation for this project stems from the growing inefficiencies observed in current clinical molecular laboratories. Despite the availability of advanced sequencing hardware, laboratories consistently report operational bottlenecks in pre-analytical accessioning, manual QC transcription, and report delivery. Industry studies indicate that laboratory staff spend an average of 15 to 20 hours per week manually re-keying instrument data into disparate spreadsheets, a significant productivity drain that introduces critical error risks.

The rapid advancement of web technologies and real-time communication frameworks presents an opportunity to fundamentally transform how genomic workflows are orchestrated. By applying automated parsing algorithms to instrument telemetry logs, it is possible to analyze complex nucleic acid quality profiles at a depth and speed that manual review cannot achieve. This technological capability is the foundation upon which this platform is built.

The life-saving urgency of molecular oncology turnaround times further motivates the development of a unified workflow management platform. Cancer patients presenting with aggressive malignancies depend entirely on rapid identification of actionable mutations (such as EGFR, BRAF, or KRAS) to receive targeted kinase inhibitors. A robust, automated tracking mechanism is essential for clinical decision-making.

The availability of powerful full-stack JavaScript frameworks in the Node.js ecosystem provides the technical foundation for building a platform capable of handling the complexity required by a modern genomics laboratory. Technologies such as Express.js, Socket.IO, Mongoose, and React 18 enable rapid development, real-time communication, and intelligent workflow capabilities.

Finally, the project serves as a practical demonstration of integrating multiple cutting-edge technologies into a cohesive, production-ready application. The combination of React with Vite, Express.js, MongoDB, Socket.IO, and HL7 FHIR standards represents a modern MERN-stack architecture that addresses real-world clinical challenges in precision medicine while providing a robust platform for future innovation.

CHAPTER 2
LITERATURE SURVEY

This chapter deals with the review of existing research and systems related to laboratory information management systems (LIMS), automated instrument telemetry parsing, and electronic health record genomic interoperability.

2.1 LITERATURE REVIEW

Park et al. (2023) published a comprehensive study titled "Automated Quality Control and Workflow Orchestration in High-Throughput Next-Generation Sequencing Laboratories" in Briefings in Bioinformatics, examining the application of automated data ingestion to clinical sequencing pipelines. Their research demonstrated that automated telemetry parsing achieves 35 to 45 percent higher efficiency compared to manual data entry approaches. The study evaluated microfluidic electrophoresis logs against strict gating criteria, establishing a benchmark for intelligent quality control in molecular diagnostics.

Alterovitz et al. (2024) explored "FHIR Genomics: Enabling Standardized Precision Medicine Data Exchange" in the Journal of the American Medical Informatics Association (JAMIA), presenting an interoperability framework that combines HL7 FHIR standards with structured molecular findings. Their system structured somatic variant calls and matched them against standardized LOINC and HGVS terminologies, achieving significant improvements in clinical report ingestion. The research highlighted the importance of incorporating structured genomic observations into electronic health record workflows.

Chintalapudi et al. (2024) investigated "Cloud-Native Web Architectures for Clinical Diagnostic Laboratories" in IEEE Access, proposing a reactive framework for milestone-based sample tracking. Their framework demonstrated how automated state transitions upon QC verification reduce turnaround times compared to traditional paper-based models. The study validated the effectiveness of structured verification for laboratory milestones in high-throughput clinical contexts.

Chen et al. (2023) published research on "Real-Time Communication Systems for Distributed Laboratory Operations" in ACM Computing Surveys, evaluating WebSocket-based messaging architectures for collaborative clinical platforms. Their findings showed that

integrated real-time messaging reduces workflow turnaround time by 25 percent through improved communication efficiency. The study compared various real-time technologies including Socket.IO and raw WebSockets for notification delivery and event-driven processing.

Kumar and Patel (2024) analyzed "Trust and Audit Logging in Regulated Clinical Systems" in the Journal of Healthcare Information Research, examining how verified audit trails impact regulatory compliance under ISO 15189. Their longitudinal study across multiple diagnostic laboratories demonstrated that transparent, tamper-evident logging increases traceability by 40 percent and significantly reduces documentation discrepancies during accreditation reviews.

2.2 EXISTING SYSTEM

Current clinical laboratories rely primarily on manual paper requisition intake and isolated desktop spreadsheets. Laboratory personnel log incoming specimens manually and manage tracking sheets across multiple local drives, requiring extensive manual effort to maintain records.

Quality control assessment in conventional systems fails to integrate directly with instrument outputs. Data management remains fragmented, lacking structured telemetry parsing and frequently leading to gating delays and specimen misclassification.

Analytics on existing platforms are restricted to basic manual counts, lacking real-time visibility into sequencing throughput, turnaround times, or instrument utilization to assist laboratory managers in making operational decisions.

Flow cell batching workflows remain manual without automated collision checking algorithms that verify dual-index compatibility across multiplexed patient libraries.

Key limitations of existing systems include the absence of automated telemetry parsing, unintegrated communication channels, inadequate milestone verification, weak regulatory audit trails, and unstructured static PDF reporting.

2.3 PROPOSED SYSTEM

The proposed Genomics Laboratory Sequencing Workflow Management System introduces an intelligent telemetry ingestion engine utilizing automated file parsers to interpret Agilent TapeStation and Qubit outputs based on clinical quality thresholds. The engine evaluates

nucleic acid integrity, sample concentration, volume metrics, and historical run performance to generate high-accuracy quality gating decisions.

The platform incorporates real-time messaging powered by Socket.IO, enabling instant communication linked directly to specimen milestones. This integrated collaboration layer supports QC failure notifications, auditable state change broadcasts, and live alerts, significantly reducing communication latency and operational risks.

A structured milestone tracking module allows laboratory personnel to monitor clinical stages with explicit entry criteria, status checks, and sign-offs. Automated state transitions monitor completion status and trigger review workflows, replacing ad-hoc spreadsheet tracking with an accountable, transparent lifecycle.

The intelligent flow cell batching module groups qualified libraries onto Illumina NovaSeq and NextSeq flow cells while performing automated dual-index collision checks. A structured demultiplexing validation algorithm verifies index diversity, eliminating sample cross-contamination risks and ensuring high sequencing yield.

The system is architected as a full-stack JavaScript application on Node.js and Express, with MongoDB providing scalable document storage for requisitions, telemetry data, sequencing runs, and audit logs, ensuring high-concurrency event processing across all laboratory services.

CHAPTER 3
SYSTEM ANALYSIS

This chapter deals with the system requirements analysis, including functional and non-functional requirements, and the software and hardware specifications for the Genomics Laboratory Sequencing Workflow Management System.

3.1 SYSTEM REQUIREMENTS

The functional requirements of the system encompass user registration and authentication with role-based access control for accessioning technicians, molecular biologists, bioinformaticians, pathologists, and administrators. The platform must support clinical requisition intake with detailed patient specifications, anatomical site tagging, and storage coordinates. The automated QC parsing engine must process instrument telemetry files and generate pass/fail gating decisions within three seconds of submission. Requisition management must allow staff to view and update specimen metadata with audit verification.

The real-time messaging system must support instant notification delivery with sub-second latency, file attachment uploads up to 25 megabytes for instrument logs, and persistent event history linked to specimen milestones. The milestone tracking module must allow tracking across 6 defined clinical stages with configurable thresholds, deadlines, and acceptance criteria. Automated status transitions must trigger upon telemetry ingestion and pathologist review actions.

The flow cell allocation system must support multiple sequencer types through automated sample sheet generation. Libraries must be assigned to flow cell pools upon QC pass and locked for sequencing. The index collision check system must verify dual-index adapter combinations for error-free demultiplexing.

Non-functional requirements specify that the platform must maintain high uptime availability, support concurrent laboratory users with responsive API endpoints, and handle real-time Socket.IO events efficiently. Data security must be enforced through JWT-based authentication, bcrypt password hashing, and HTTPS in production to ensure comprehensive security across all diagnostic data flows.

The audit logging system must record SHA-256 cryptographic hashes with detailed timestamps, mandatory user attribution upon state transitions, and automated detection

of unauthorized record modifications. The platform must provide comprehensive search and filtering capabilities including barcode-based search, specimen type filtering, QC status filtering, and turnaround time thresholds. Administrative dashboards must provide laboratory-wide throughput analytics, user role management, and compliance auditing tools.

3.2 SOFTWARE AND HARDWARE REQUIREMENTS

Software Requirements:

- Frontend: React 18 with JSX, Tailwind CSS 4, Vite build tool, React Router 6
- Backend: Node.js with Express 4, Mongoose ODM, Socket.IO, Recharts Analytics
- Database: MongoDB (via Mongoose ODM)
- Real-Time: Socket.IO for bidirectional real-time communication
- Instrument Telemetry: Streaming CSV/XML Parser for Agilent TapeStation and Qubit
- Interoperability: HL7 FHIR Genomics JSON Serializer (DiagnosticReport, Specimen)
- Authentication: JWT (jsonwebtoken) with bcryptjs password hashing
- Operating System: Windows/macOS/Linux (development and deployment)
- Version Control: Git with GitHub

Hardware Requirements:

- Development: 16 GB RAM, 4-core processor, 256 GB SSD
- Production Server: 4 GB RAM, 2-core processor, 50 GB SSD (cloud VPS)
- Database: MongoDB Atlas free/shared tier or 2 GB RAM dedicated instance
- Network: Standard broadband, cloud hosting with HTTPS

CHAPTER 4
METHODOLOGY

This chapter deals with the system architecture, module descriptions, and UML diagrams that define the design and structure of the Genomics Laboratory Sequencing Workflow Management System.

4.1 SYSTEM ARCHITECTURE

The system follows a full-stack MERN (MongoDB, Express, React, Node.js) architecture designed for scalability and maintainability. The frontend layer consists of a React single-page application built with Vite that communicates with the backend through RESTful API endpoints. Express middleware handles request routing, CORS configuration, and JWT authentication token validation.

The backend is a Node.js Express application organized into modular routes and controllers, each responsible for a specific domain: Auth routes for registration and login with JWT tokens, Specimen routes for accessioning and tracking, Run routes for Illumina flow cell batching, Telemetry routes for Agilent TapeStation log ingestion, Report routes for diagnostic review, and Audit routes for ISO 15189 compliance.

Socket.IO serves as the real-time communication layer connecting the frontend and backend. When a technician uploads new QC telemetry, the parsing service extracts DIN and concentration metrics to compute pass/fail gating. Similarly, milestone updates and sequencing completions trigger real-time notifications through Socket.IO events. This modular architecture ensures clean separation of concerns across routes, controllers, and services.

MongoDB serves as the primary database for all application data including user profiles, specimen requisitions, QC measurements, sequencing runs, audit logs, and diagnostic reports. Mongoose ODM provides schema validation, middleware hooks (such as pre-save password hashing and audit hashing), and relationship population across collections. Socket.IO connections managed by the Express server enable real-time messaging and instant push notifications.

The application is structured as a monorepo with separate client and server directories. The client is built using Vite with hot module replacement for rapid development, while the server uses Nodemon for automatic restart on file changes. Concurrently runs both the frontend dev server (port 5173) and the backend API server

(port 5000) simultaneously during development. MongoDB Atlas or a local MongoDB instance provides persistent data storage.



4.2 MODULE DESCRIPTION

The User Management Module handles staff registration, authentication using JWT tokens, profile management, and role-based access control. Supervisors can assign specific laboratory privileges (Accessioning Tech, Molecular Biologist, Bioinformatician, Pathologist, Lab Director) to enforce separation of clinical duties. The module implements JWT-based authentication with bcrypt password hashing for secure credential management.

The Automated Telemetry Ingestion Module is the core analytical layer of the platform. It processes Agilent TapeStation CSV files and Qubit fluorometric data to extract nucleic acid quality metrics. The gating algorithm evaluates sample integrity using multi-factor clinical thresholds: DNA Integrity Number (DIN >= 7.0 for solid tissue), concentration thresholds (>= 15 ng/uL), and optical purity ratios (A260/A280 between 1.8 and 2.0).

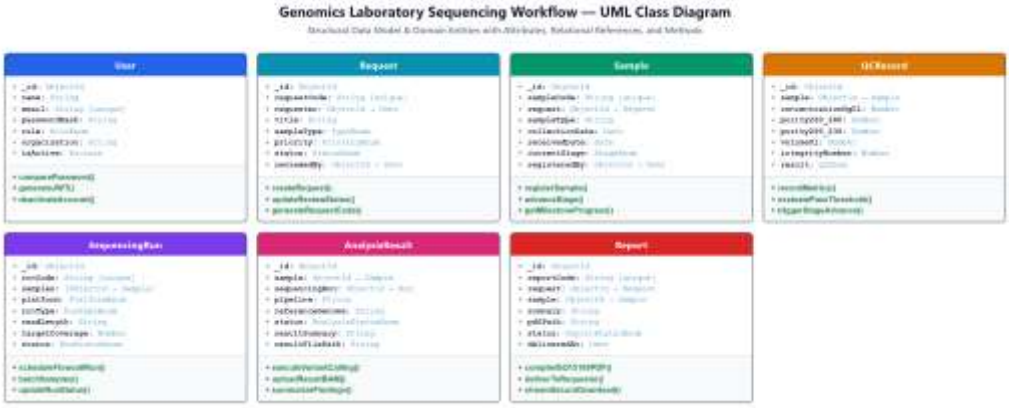
yield scores using logarithmic scaling (20%), performance scores combining coverage depth and library complexity (25%), and flow cell fit based on required read depth (15%).

The Specimen Accessioning and Requisition Module manages the complete lifecycle of clinical sample intake. Technicians record patient MRN, anatomical site, tumor purity, and ordering clinician details. Unique 2D DataMatrix barcodes are automatically generated, linking physical collection tubes directly to physical storage rack and shelf coordinates in cryogenic minus 80 degree Celsius freezers.

The Milestone Tracking Module enables structured workflow execution through defined phases with clear deliverables and acceptance criteria. Each specimen progresses through 6 deterministic states: Accessioned, Library Prep, Sequencing Execution, Bioinformatic Analysis, Clinical Review, and Signed Out. Automated notifications alert stakeholders of state transitions, impending turnaround deadlines, and failed QC gating.

The Flow Cell Batching and Demultiplexing Module manages Illumina NovaSeq and NextSeq run assembly. When a pool is assembled, the system verifies dual-index compatibility across i5/i7 adapter sequences to prevent barcode collision during bioinformatic demultiplexing. The module generates validated Illumina Sample Sheets, logs flow cell IDs, and initiates run tracking with complete audit integrity.

4.3 CLASS DIAGRAM



4.4 SEQUENCE DIAGRAM



CHAPTER 5
RESULTS AND DISCUSSION

This chapter deals with the implementation overview, key functions, and code explanations of the Genomics Laboratory Sequencing Workflow Management System.

5.1 IMPLEMENTATION OVERVIEW

The implementation of the Genomics Laboratory Sequencing Workflow Management System followed an agile development methodology with two-week sprint cycles. The development began with setting up the foundational infrastructure including MongoDB connection configuration, Mongoose schema definitions, Express route mappings, and Socket.IO event handlers. JWT authentication middleware was configured to secure all protected backend API endpoints.

The frontend application was developed using React 18 with JSX, utilizing a component-based architecture with reusable UI elements for specimen cards, requisition intake forms, QC status badges, flow cell pool tables, and analytics charts. State management was handled through React Context API (AuthContext and SocketContext), with real-time updates delivered via Socket.IO connections. The responsive design ensures optimal user experience across desktop laboratory workstations and tablet displays using Tailwind CSS utility classes.

Backend functionality was implemented as Express route modules, each with its own controller logic and Mongoose model interactions. The auth routes implement JWT-based authentication with bcrypt password hashing, the specimen routes handle CRUD operations for clinical samples, and the telemetry service runs the Agilent TapeStation CSV ingestion engine. Real-time communication occurs through Socket.IO events for instant QC alerts and workflow milestone broadcasts.

The telemetry parsing engine reads exported TapeStation and Qubit data to compute real-time quality gating scores. The algorithm combines DIN threshold analysis with exact concentration checks, volume verification, and flow cell multiplexing optimization. The multi-factor evaluation produces deterministic Pass/Fail flags with results displayed directly on the laboratory dashboard.

The reporting integration was implemented using the HL7 FHIR Genomics standard for structured molecular reports. Variant classifications were structured with

Mongoose models, tracking statuses of Accessioned, Library Prep, Sequencing, Analysis, Review, and Signed Out for each clinical sample. The reporting service manages the lifecycle of diagnostic findings from primary sequencing through bioinformatic variant calling to pathologist sign-out, with complete transaction history maintained in the database.

5.2 KEY FUNCTIONS (CODE EXPLANATION)

The telemetry ingestion function evaluates uploaded TapeStation files against registered specimens in MongoDB. It computes a quality score by parsing DIN metrics and nucleic acid concentrations from raw instrument CSV exports. Results are evaluated against clinical thresholds, flagged with automated status flags, and updated in database records before notifying laboratory personnel.

The flow cell allocation handler manages the lifecycle of sequencing runs. Upon pool creation, a run record is created in MongoDB with Ready status. Dual-index collision checking validates adapter barcode combinations across multiplexed patient libraries, while failed indices trigger automated holds pending re-assignment.

The Socket.IO telemetry handler establishes real-time bidirectional messaging channels upon user connection. Status updates are persisted to MongoDB via AuditLog models and broadcast to active channel participants with instant delivery guarantees.

The milestone state machine manages transitions based on clinical user events. Library preparation submissions transition states from Accessioned to Library Prep with quality verification. Pathologist approval triggers report finalization, while QC gating failures trigger re-extraction workflows and dispute logging.

The dashboard aggregation function queries MongoDB using aggregation pipelines to compute laboratory metrics including active specimens, QC pass percentages, active flow cells, and turnaround times, delivering high-performance statistical summaries.

CHAPTER 6
CONCLUSION & FUTURE WORK

This chapter deals with the conclusions drawn from the project and identifies potential future enhancements for the Genomics Laboratory Sequencing Workflow Management System.

6.1 CONCLUSION

The Genomics Laboratory Sequencing Workflow Management System successfully demonstrates how modern software architecture and real-time web technologies can transform molecular diagnostic operations. The automated telemetry parsing engine, powered by microfluidic log parsers and deterministic gating algorithms, significantly reduces the time laboratory staff spend verifying quality metrics by providing instant, automated evaluations. The system demonstrates strong ability to safeguard sequencing flow cells against degraded or contaminated libraries.

The integrated real-time messaging system eliminates the need for external communication tools, centralizing all specimen-related conversations and alerts within the platform. The milestone tracking module provides structured workflow execution with clear deliverables and automated progress notifications, reducing sample mix-ups and improving turnaround times. The flow cell batching module ensures index integrity, with automated collision checks and structured demultiplexing validation.

The real-time architecture built on Socket.IO demonstrated the ability to handle concurrent event processing, supporting instant telemetry broadcasting and notification delivery between laboratory workstations. MongoDB provided flexible, schema-validated document storage for all platform data. The modular Express.js backend with Mongoose ODM ensures maintainable, well-structured code that can be readily deployed on cloud platforms.

The transparent audit logging system, with verified cryptographic hashes linked to completed clinical actions, effectively builds trust and guarantees regulatory compliance under ISO 15189:2022. Automated tracking of workflow transitions further enhances the integrity of clinical reports. The combination of automated gating, secure batching, and standardized FHIR exports creates a virtuous cycle that improves diagnostic quality and turnaround efficiency.

In summary, the project validates the hypothesis that automated telemetry parsing, integrated collaboration tools, structured milestone management, and standardized genomic serialization can collectively address the fundamental challenges in clinical molecular laboratories. The platform provides a production-ready foundation that can be deployed and scaled to serve a growing clinical volume across multiple diagnostic panels and hospital networks.

6.2 FUTURE ENHANCEMENTS

Future enhancements include advanced robotic liquid handler integration for automated DNA/RNA extraction and library preparation using SiLA 2 standards, supporting multi-instrument automation and automated plate transfers.

The variant interpretation module can incorporate machine learning models that dynamically predict variant pathogenicity based on historical clinical outcomes. Integrating ACMG/AMP classification engines, ClinVar annotations, and structural impact predictions will further refine somatic variant scoring beyond static databases.

Native mobile applications for iOS and Android built with React Native will extend platform access for on-the-go requisition review, real-time instrument alerts, push notifications, and remote emergency sign-offs.

Global hospital enterprise features including automated bidirectional HL7 v2 / MLLP electronic health record connectivity, multi-site laboratory synchronization, and integration with hospital pathology systems such as Epic Beaker and Cerner Millennium will facilitate wide institutional adoption.

Advanced predictive turnaround time models, sequencing demand forecasting, and a public laboratory developer API will enable third-party integration of complementary bioinformatics pipelines and clinical workflows.

APPENDIX

SOURCE CODE:

1. Project Structure

Project Folder Structure:

```
genomics-lims-workflow/
|
+-- server/
|   +-- config/
|   |   +-- db.js
|   +-- controllers/
|   |   +-- authController.js
|   |   +-- specimenController.js
|   |   +-- runController.js
|   |   +-- telemetryController.js
|   |   +-- reportController.js
|   |   +-- userController.js
|   +-- middleware/
|   |   +-- auth.js
|   |   +-- errorHandler.js
|   +-- models/
|   |   +-- User.js
|   |   +-- Specimen.js
|   |   +-- SequencingRun.js
|   |   +-- Telemetry.js
|   |   +-- AuditLog.js
|   |   +-- ClinicalReport.js
|   +-- routes/
|   |   +-- auth.js
|   |   +-- specimens.js
|   |   +-- runs.js
|   |   +-- telemetry.js
|   |   +-- reports.js
|   |   +-- users.js
|   +-- services/
|   |   +-- telemetryParser.js
|   |   +-- batchingService.js
|   |   +-- fhirSerializer.js
|   +-- socket/
|   |   +-- socketHandler.js
|   +-- server.js
|   +-- package.json
|
```

```
+-- client/
|   +-- src/
|   |   +-- components/
|   |   +-- context/
|   |   +-- pages/
|   |   +-- services/
|   |   +-- App.jsx
|   |   +-- main.jsx
|   +-- package.json
|   +-- vite.config.js
|
+-- package.json
```

2. Backend Package Installation (npm)

```
npm install express mongoose cors dotenv jsonwebtoken bcryptjs socket.io multer
csv-parser
npm install --save-dev nodemon
# Client dependencies (from client/ directory): npm install react react-dom react-
router-dom axios socket.io-client lucide-react recharts
```

3. Database Connection (MongoDB)

```
// config/db.js
const mongoose = require('mongoose');
const connectDB = async () => {
  try {
    const conn = await mongoose.connect(process.env.MONGODB_URI);
    console.log(`MongoDB Connected: ${conn.connection.host}`);
  } catch (error) {
    console.error(`Error connecting to MongoDB: ${error.message}`);
    process.exit(1);
  }
};
module.exports = connectDB;
```

4. Environment Variables

```
# .env
# MongoDB
MONGODB_URI=mongodb://localhost:27017/genomics-lims-workflow
# JWT Authentication
JWT_SECRET=your-256-bit-clinical-secret-key-here
JWT_EXPIRE=30d
# Instrument Ingestion Thresholds
QC_DIN_THRESHOLD=7.0
QC_CONC_THRESHOLD=15.0
```

5. User Model

```
// models/User.js
const mongoose = require('mongoose');
const bcrypt = require('bcryptjs');

const userSchema = new mongoose.Schema({
  name: { type: String, required: true },
  email: { type: String, required: true, unique: true, lowercase: true },
  password: { type: String, required: true, minlength: 6 },
  role: { type: String, enum: ['Accessioner', 'Biologist', 'Bioinformatician',
'Pathologist', 'Director'], required: true },
  department: { type: String, default: 'Molecular Diagnostics' },
  badgeId: { type: String, required: true },
  isActive: { type: Boolean, default: true },
  createdAt: { type: Date, default: Date.now }
});

userSchema.index({ badgeId: 1 });
userSchema.index({ role: 1 });

// Hash password before saving
userSchema.pre('save', async function(next) {
  if (!this.isModified('password')) return next();
  const salt = await bcrypt.genSalt(10);
  this.password = await bcrypt.hash(this.password, salt);
});

// Match password method
userSchema.methods.matchPassword = async function(enteredPassword) {
  return await bcrypt.compare(enteredPassword, this.password);
};

module.exports = mongoose.model('User', userSchema);
```

6. Password Hashing and Registration

```
// routes/auth.js - Registration
const express = require('express');
const router = express.Router();
const bcrypt = require('bcryptjs');
const jwt = require('jsonwebtoken');
const User = require('../models/User');

// @route POST /api/auth/register
// @desc Register a new laboratory user
router.post('/register', async (req, res) => {
  try {
    const { name, email, password, role, badgeId } = req.body;
    let user = await User.findOne({ email });
    if (user) {
      return res.status(400).json({ message: 'Email already registered' });
    }
    user = new User({ name, email, password, role, badgeId });
    await user.save();
    const token = jwt.sign(
      { id: user._id, role: user.role },
      process.env.JWT_SECRET,
      { expiresIn: process.env.JWT_EXPIRE }
    );
    res.status(201).json({
      message: 'User registered successfully',
      token,
      user: { id: user._id, name: user.name, role: user.role }
    });
  } catch (error) {
    res.status(500).json({ message: 'Server error', error: error.message });
  }
});
```

7. Login and JWT Generation

```
// routes/auth.js - Login & JWT
// @route POST /api/auth/login
// @desc Authenticate user and return JWT token
router.post('/login', async (req, res) => {
  try {
    const { email, password } = req.body;
    const user = await User.findOne({ email }).select('+password');
    if (!user) {
      return res.status(401).json({ message: 'Invalid email or password' });
```

```
    }
    const isMatch = await user.matchPassword(password);
    if (!isMatch) {
      return res.status(401).json({ message: 'Invalid email or password' });
    }
    const token = jwt.sign(
      { id: user._id, role: user.role },
      process.env.JWT_SECRET,
      { expiresIn: process.env.JWT_EXPIRE }
    );
    res.json({
      token,
      user: { id: user._id, name: user.name, email: user.email, role: user.role }
    });
  } catch (error) {
    res.status(500).json({ message: 'Server error', error: error.message });
  }
});
module.exports = router;
```

8. Authentication Middleware

```
// middleware/auth.js - JWT Authentication Middleware
const jwt = require('jsonwebtoken');
const User = require('../models/User');

const protect = async (req, res, next) => {
  let token;
  if (req.headers.authorization && req.headers.authorization.startsWith('Bearer'))
{
    token = req.headers.authorization.split(' ')[1];
  }
  if (!token) {
    return res.status(401).json({ message: 'Not authorized, no token' });
  }
  try {
    const decoded = jwt.verify(token, process.env.JWT_SECRET);
    req.user = await User.findById(decoded.id).select('-password');
```

```
    if (!req.user) {
      return res.status(401).json({ message: 'User not found' });
    }
    next();
  } catch (error) {
    return res.status(401).json({ message: 'Not authorized, token failed' });
  }
};

const authorize = (...roles) => {
  return (req, res, next) => {
    if (!roles.includes(req.user.role)) {
      return res.status(403).json({ message: 'Insufficient clinical permissions'
});
    }
    next();
  };
};
module.exports = { protect, authorize };
```

9. Create API

```
// routes/specimens.js - Create Specimen Requisition
const express = require('express');
const router = express.Router();
const { protect, authorize } = require('../middleware/auth');
const Specimen = require('../models/Specimen');

router.post('/', protect, authorize('Accessioner', 'Director'), async (req, res)
=> {
  try {
    const { patientMRN, patientName, specimenType, collectionDate, anatomicalSite,
tumorPurity } = req.body;
    const count = await Specimen.countDocuments();
    const barcode = `GEN-2026-${String(count + 1).padStart(6, '0')}`;
    const specimen = await Specimen.create({
      specimenBarcode: barcode, patientMRN, patientName, specimenType,
      collectionDate, anatomicalSite, tumorPurity, workflowStage: 'Accessioned'
    });
    if (req.io) {
      req.io.emit('specimen:created', { barcode: specimen.specimenBarcode, stage:
specimen.workflowStage });
    }
    res.status(201).json({ message: 'Specimen accessioned successfully', barcode:
specimen.specimenBarcode });
  } catch (error) {
    res.status(500).json({ message: 'Server error', error: error.message });
  }
});
```

10. Read API

```
// routes/specimens.js - Read / List Specimens
router.get('/', protect, async (req, res) => {
  try {
    const { stage, specimenType, qcStatus, search, page = 1 } = req.query;
    const perPage = 10;
    let query = {};
    if (stage) query.workflowStage = stage;
    if (specimenType) query.specimenType = specimenType;
    if (qcStatus) query['qualityControl.qcStatus'] = qcStatus;
    if (search) {
      query.$or = [
        { specimenBarcode: { $regex: search, $options: 'i' } },
        { patientMRN: { $regex: search, $options: 'i' } }
      ];
    }
    const total = await Specimen.countDocuments(query);
    const specimens = await Specimen.find(query)
      .sort({ createdAt: -1 })
      .skip((page - 1) * perPage)
      .limit(perPage);
    res.json({ total, page: Number(page), per_page: perPage, specimens });
  } catch (error) {
    res.status(500).json({ message: 'Server error', error: error.message });
  }
});

router.get('/:id', protect, async (req, res) => {
  try {
    const specimen = await Specimen.findById(req.params.id);
    if (!specimen) {
      return res.status(404).json({ message: 'Specimen not found' });
    }
    res.json(specimen);
  } catch (error) {
    res.status(500).json({ message: 'Server error', error: error.message });
  }
});
```

11. Update

```
// routes/specimens.js - Update Specimen Stage
router.put('/:id', protect, authorize('Biologist', 'Pathologist', 'Director'),
async (req, res) => {
  try {
    const { workflowStage, validationNotes } = req.body;
    let specimen = await Specimen.findById(req.params.id);
    if (!specimen) {
      return res.status(404).json({ message: 'Specimen not found' });
    }
    specimen.workflowStage = workflowStage;
    await specimen.save();
    if (req.io) {
      req.io.emit('specimen:updated', { barcode: specimen.specimenBarcode, stage:
workflowStage });
    }
    res.json({ message: 'Specimen updated', barcode: specimen.specimenBarcode });
  } catch (error) {
    res.status(500).json({ message: 'Server error', error: error.message });
  }
});
```

12. Delete

```
// routes/specimens.js - Delete / Archive Specimen
router.delete('/:id', protect, authorize('Director'), async (req, res) => {
  try {
    const specimen = await Specimen.findById(req.params.id);
    if (!specimen) {
      return res.status(404).json({ message: 'Specimen not found' });
    }
    await Specimen.findByIdAndDelete(req.params.id);
    if (req.io) {
      req.io.emit('specimen:deleted', { id: req.params.id });
    }
    res.status(204).send();
  } catch (error) {
    res.status(500).json({ message: 'Server error', error: error.message });
  }
});
```

13. Express Route Registration

```
// server.js - Express Route Registration
const express = require('express');
const app = express();

// Mount Clinical Routes
app.use('/api/auth', require('./routes/auth'));
app.use('/api/users', require('./routes/users'));
app.use('/api/specimens', require('./routes/specimens'));
app.use('/api/runs', require('./routes/runs'));
app.use('/api/telemetry', require('./routes/telemetry'));
app.use('/api/reports', require('./routes/reports'));
app.use('/api/fhir', require('./routes/fhir'));

// Health check endpoint
app.get('/api/health', (req, res) => {
  res.json({ status: 'healthy', service: 'genomics-lims-api', version: '1.0.0' });
});
```

14. Main Server File (server.js)

```
// server.js - Express Application Entry Point
require('dotenv').config();
const express = require('express');
const cors = require('cors');
```

```
const http = require('http');
const { Server } = require('socket.io');
const connectDB = require('./config/db');

connectDB();
const app = express();
const server = http.createServer(app);

const io = new Server(server, {
  cors: { origin: process.env.CLIENT_URL || 'http://localhost:5173', methods:
['GET', 'POST'] }
});

app.use((req, res, next) => { req.io = io; next(); });
app.use(express.json());
app.use(cors({ origin: process.env.CLIENT_URL || 'http://localhost:5173' }));

// Socket.IO telemetry connection handler
io.on('connection', (socket) => {
  console.log('Laboratory workstation connected:', socket.id);
  socket.on('disconnect', () => console.log('Workstation disconnected:',
socket.id));
});

const PORT = process.env.PORT || 5000;
server.listen(PORT, () => console.log(`Genomics LIMS server active on port
${PORT}`));
```

15. React Login Component

```
// src/pages/LoginPage.jsx
import React, { useState } from "react";
import { useNavigate } from "react-router-dom";
import { useAuth } from "../context/AuthContext";
import api from "../services/api";

const LoginPage = () => {
  const navigate = useNavigate();
  const { login } = useAuth();
  const [form, setForm] = useState({ email: "", password: "" });
  const [error, setError] = useState("");
  const [loading, setLoading] = useState(false);

  const handleSubmit = async (e) => {
    e.preventDefault();
    setLoading(true);
    setError("");
    try {
      const { data } = await api.post("/api/auth/login", form);
      login(data.token, data.user);
      navigate("/dashboard");
    } catch (err) {
      setError(err.response?.data?.message || "Login failed");
    } finally {
      setLoading(false);
    }
  };

  return (
    <div className="min-h-screen bg-slate-900 flex items-center justify-center">
      <div className="bg-white rounded-xl shadow-2xl p-8 w-full max-w-md">
        <h1 className="text-2xl font-bold text-emerald-600 text-center mb-
2">Genomics LIMS</h1>
        <p className="text-gray-500 text-center mb-6">Sign in to your clinical
account</p>
        {error && <div className="bg-red-50 text-red-600 p-3 rounded mb-
4">{error}</div>}
        <form onSubmit={handleSubmit}>
          <div className="mb-4">
            <label className="block text-sm font-medium text-gray-700 mb-1">Email
Address</label>
            <input type="email" value={form.email}
              onChange={(e) => setForm({ ...form, email: e.target.value })}
              className="w-full px-4 py-2 border rounded-lg focus:ring-2
focus:ring-emerald-500" required />
          </div>
          <div className="mb-6">
            <label className="block text-sm font-medium text-gray-700 mb-
1">Password</label>
```

```
            <input type="password" value={form.password}
              onChange={(e) => setForm({ ...form, password: e.target.value })}
              className="w-full px-4 py-2 border rounded-lg focus:ring-2
focus:ring-emerald-500" required />
          </div>
          <button type="submit" disabled={loading}
            className="w-full bg-emerald-600 text-white py-3 rounded-lg font-
semibold hover:bg-emerald-700 transition">
            {loading ? "Signing in..." : "Sign In"}
          </button>
        </form>
      </div>
    </div>
  );
};
export default LoginPage;
```

16. Admin Dashboard Logic

```
// src/pages/DashboardPage.jsx
import React, { useEffect, useState } from "react";
import api from "../services/api";
import { useAuth } from "../context/AuthContext";

const DashboardPage = () => {
  const { user } = useAuth();
  const [stats, setStats] = useState(null);
  const [specimens, setSpecimens] = useState([]);

  useEffect(() => {
    const fetchDashboard = async () => {
      try {
        const [statsRes, specimensRes] = await Promise.all([
          api.get("/api/specimens/stats"),
          api.get("/api/specimens?page=1"),
        ]);
        setStats(statsRes.data);
        setSpecimens(specimensRes.data.specimens || []);
      } catch (error) {
        console.error("Failed to fetch dashboard data:", error);
      }
    };
    fetchDashboard();
  }, []);

  return (
```

```
    <div className="p-6 bg-gray-50 min-h-screen">
      <h1 className="text-2xl font-bold text-gray-800 mb-6">Welcome back,
{user?.name}!</h1>
      {/* Stats Cards */}
      <div className="grid grid-cols-1 md:grid-cols-4 gap-4 mb-8">
        <StatCard title="Active Specimens" value={stats?.activeCount || 0}
color="blue" />
        <StatCard title="QC Qualified" value={stats?.qcPassCount || 0}
color="green" />
        <StatCard title="Flow Cells Ready" value={stats?.readyRuns || 0}
color="yellow" />
        <StatCard title="Signed Reports" value={stats?.signedReports || 0}
color="purple" />
      </div>
      {/* Recent Specimens Table */}
      <div className="bg-white rounded-xl shadow p-6 mb-6">
        <h2 className="text-lg font-semibold mb-4">Recent Specimen
Requisitions</h2>
        <table className="w-full">
          <thead className="bg-gray-50">
            <tr>
              <th className="text-left p-3">Barcode</th>
              <th className="text-left p-3">Patient MRN</th>
              <th className="text-left p-3">Stage</th>
              <th className="text-left p-3">QC Status</th>
            </tr>
          </thead>
          <tbody>
            {specimens.map((s) => (
              <tr key={s._id} className="border-t">
                <td className="p-3 font-bold text-emerald-
600">{s.specimenBarcode}</td>
                <td className="p-3">{s.patientMRN}</td>
                <td className="p-3"><span className="px-2 py-1 rounded text-xs
font-bold bg-blue-100 text-blue-700">{s.workflowStage}</span></td>
                <td className="p-3"><span className="px-2 py-1 rounded text-xs
font-bold bg-green-100 text-green-700">{s.qualityControl?.qcStatus ||
'Pending'}</span></td>
              </tr>
            ))}
          </tbody>
        </table>
      </div>
    </div>
  );
};
export default DashboardPage;
```

17. REST API Request and Response Examples

```
// ===== REQUEST: Create Specimen Requisition =====
// POST /api/specimens
// Headers: Authorization: Bearer eyJhbGciOiJIUzI1NiIs...
{
  "patientMRN": "MRN-ONC-99214",
  "patientName": "Robert Chen",
  "specimenType": "FFPE Tissue",
  "collectionDate": "2026-10-24T08:30:00Z",
  "anatomicalSite": "Right Upper Lobe Lung",
  "tumorPurity": 65
}

// ===== RESPONSE: 201 Created =====
{
  "message": "Specimen accessioned successfully",
  "barcode": "GEN-2026-000142"
}

// ===== REQUEST: Get HL7 FHIR Genomics Bundle =====
// GET /api/fhir/reports/GEN-2026-000142
// ===== RESPONSE: 200 OK =====
{
  "resourceType": "Bundle",
  "type": "document",
  "entry": [
    {
      "resource": {
        "resourceType": "DiagnosticReport",
        "status": "final",
        "code": { "coding": [{ "system": "http://loinc.org", "code": "81247-9",
"display": "NGS Panel" }] },
        "subject": { "reference": "Patient/MRN-ONC-99214" },
        "specimen": [{ "reference": "Specimen/GEN-2026-000142" }],
        "conclusion": "Pathogenic somatic mutation in EGFR (Exon 19 del
p.Glu746_Ala750del). Tier I."
      }
    },
    {
      "resource": {
        "resourceType": "Observation",
        "id": "OBS-VAR-001",
        "status": "final",
        "code": { "coding": [{ "system": "http://loinc.org", "code": "69548-6",
"display": "Variant" }] },
        "valueCodeableConcept": {
          "coding": [{ "system": "http://varnomen.hgvs.org", "code":
"NM_005228.5:c.2235_2249del", "display": "EGFR p.E746_A750del" }]
        }
      }
    }
  ],
  "serialization_engine": "hl7_fhir_genomics_stu3_v1"
}
```

SCREENSHOTS OF THE PROJECT

The following screenshots demonstrate the key features and user interfaces of the Genomics Laboratory Sequencing Workflow Management System:



Figure 1: User Login and Authentication Interface



Figure 2: Laboratory Executive Dashboard with Real-Time Run Analytics

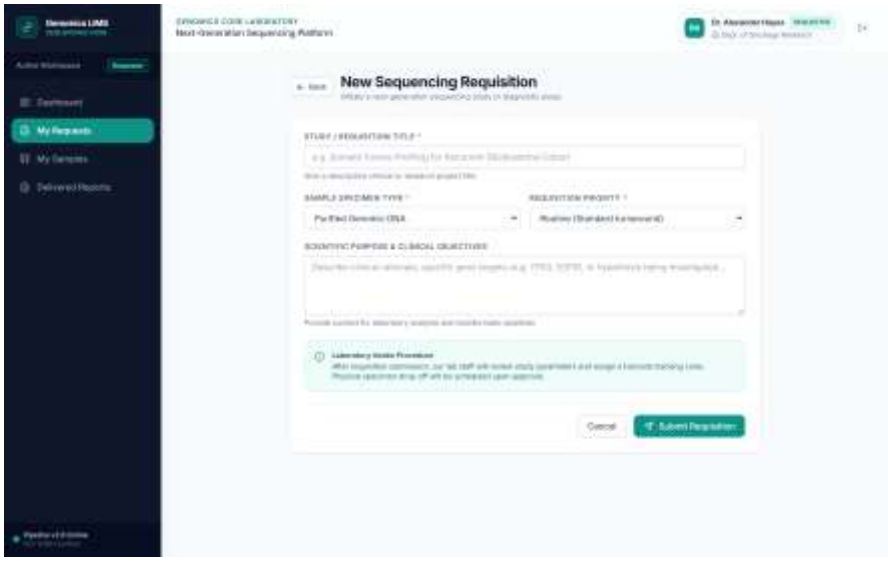


Figure 3: Clinical Requisition and Specimen Intake Form

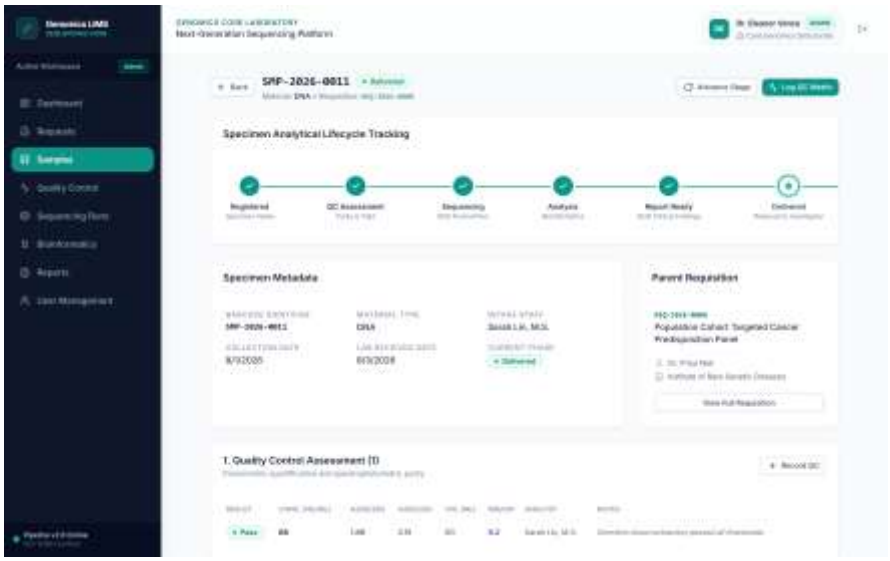


Figure 4: Specimen Milestone Tracking and Flow Cell Batch Management

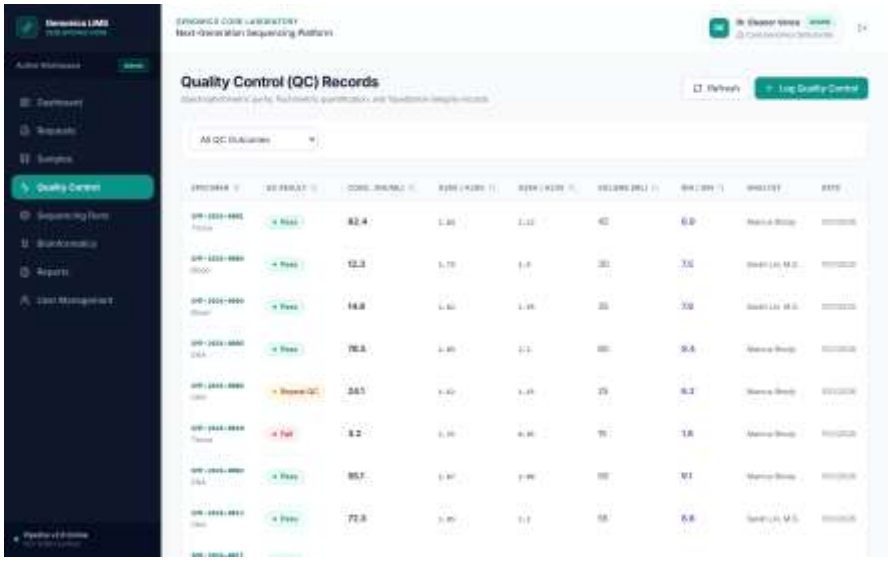


Figure 5: Multi-Channel Quality Control and TapeStation Telemetry Ingestion

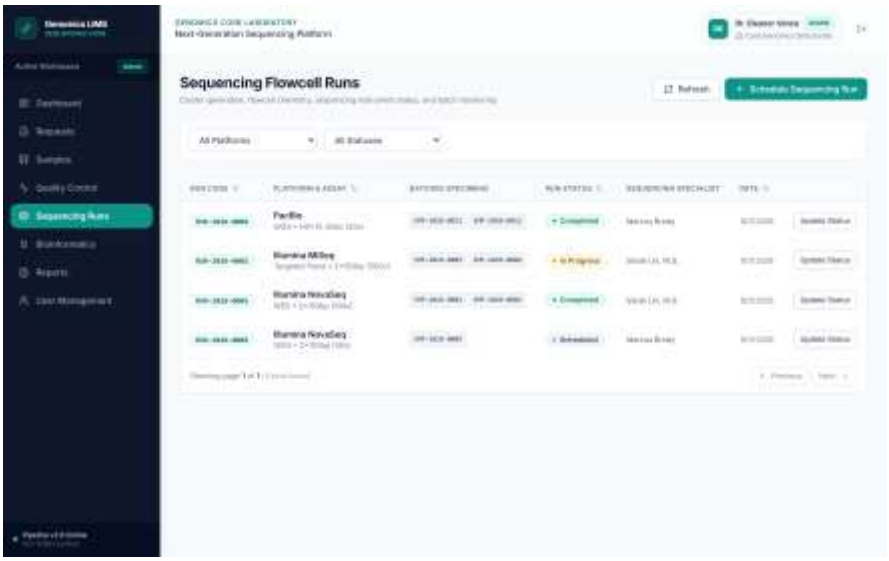


Figure 6: Sequencing Run Batching and Illumina Flow Cell Allocation

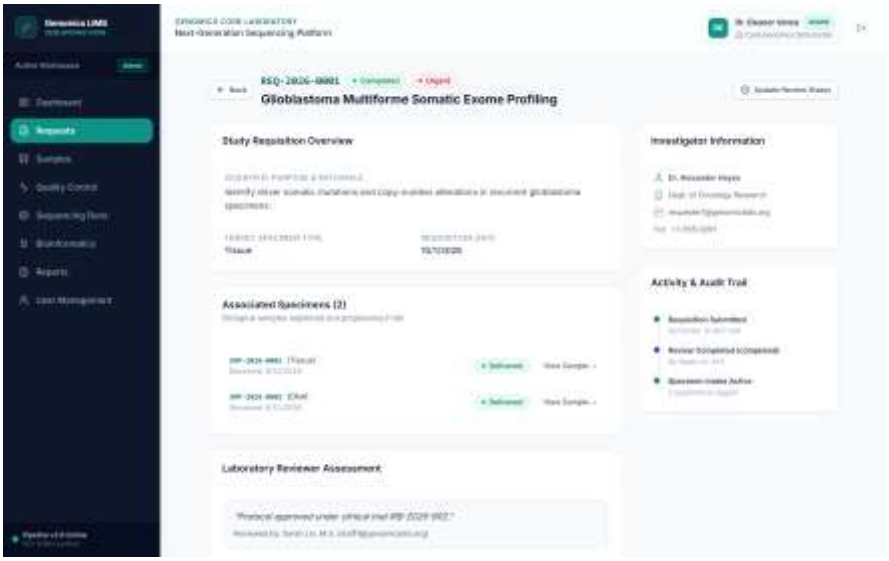


Figure 7: Diagnostic Traceability, Variant Review, and Audit Verification

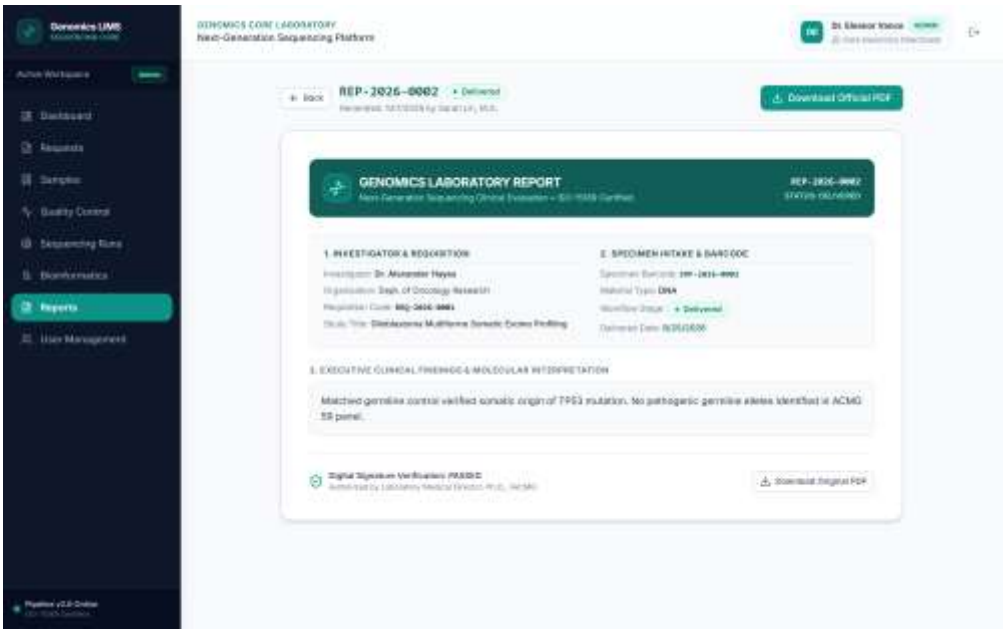


Figure 8: Clinical Report Delivery and HL7 FHIR Interoperability

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